American Computer Science League

2024-2025 ● Contest 3: Short Problems Solutions ● Intermediate Division

1. Boolean Algebra

$A(\overline{B} + C) + \overline{AB} + C = A\overline{B} + AC + \overline{A} + \overline{B} + C$
$= \overline{B}(A + 1) + C(A + 1) + \overline{A}$
$= \overline{B} + C + \overline{A}$
The only time this expression is FALSE is when all terms are 0, which is (1, 1, 0). So 7 triples make it TRUE.

E. 7

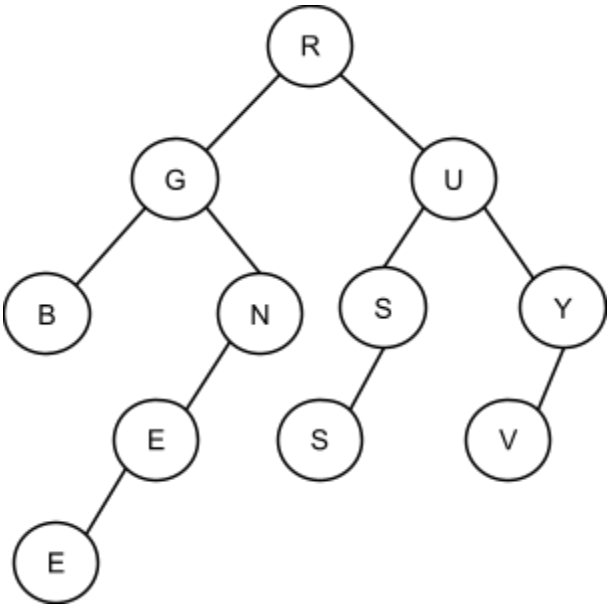
2. Boolean Algebra

$\overline{A}B(\overline{C + A}) + A\,B(\overline{\overline{A} + C})$
$= \overline{A}\,B\,\overline{C}\,\overline{A} + A\,B\,A\,\overline{C}$
$= \overline{A}\,B\,\overline{C} + A\,B\,\overline{C}$
$= B\,\overline{C}\,(\overline{A} + A)$
$= B\,\overline{C}$

D. $B\overline{C}$

3. Data Structures

The binary search tree for **RUGBYSEVENS** is:



The nodes at depth 2 from left to right are: B, N, S, Y

B. B, N, S, Y

4. Data Structures The stack from top to bottom is:' A, AR, ARC, AR, ARH, AR, A, AE, AER, AERY, AER, AE. The next item popped would be E.	C. E
5. FSAs a. 01111101 satisfies it b. 011101 satisfies it c. 0111111 must contain a 0 before the final 1 d. 0111001 satisfies it e. 0110 must end in 1 f. 011110001 satisfies it	D. 4
6. Regular Expressions [J-PT-V]*[^c^f-m]*[2468] a. Jjp2 fails at j in [^c^f-m] b. KULaazzz24 fails at 4 in [2468] c. L6 satisfies it d. TTOOB8 satisfies it e. dstxyz4 satisfies it f. 4 satisfies it	C. b, c, d, e, f